From Configurations to Claims: An Evidence Audit of Urban EV Charging Forecasting

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Abstract

Execution completion is often treated as a proxy for reproducibility, while a statistically positive comparison is often treated as sufficient reason to continue model development. We audit both assumptions in a case study of urban electric-vehicle charging forecasting and extract a reusable claim-control framework linking configuration identity, forecast validity, engineering qualification, and protected-data use. In a frozen inventory of 1,224 UrbanEV configurations, 865 have completed artifacts but only 156 satisfy a joint exact-protocol-and-numerical criterion. A sequential UrbanEV evaluation shows why later evidence types must remain distinct: strict prequential fusion of a locally audited CAPER expert and compact TimeXer lowers internal macro RMSE by 3.178% relative to the strongest evaluated single model, but increases batch-1 median synchronous latency by 48.86% on the frozen device. Subsequent routing, foundation-teacher, and residual-distillation candidates either deteriorate or remain below their internally time-stamped, project-specific progression gates. On a separate Paris development-only panel, the strict teacher is consistently positive yet its 0.847% point gain remains below the frozen 1% rule. Paris formal and protected target-bearing sources are confined to a procedurally isolated evidence vault and contribute no analytical result. The resulting taxonomy, claim grammar, fail-closed state machine, and artifact ledger show how a forecasting project can preserve partial and negative evidence without promoting it to protected-test, deployment, or state-of-the-art claims.

Keywords: electric-vehicle charging; forecasting; reproducibility; prequential evaluation; experiment audit; latency; stopping rules

1 Introduction

An experiment can finish without reproducing the claim it appears to test. A run may emit a checkpoint and a metric even when the source protocol does not uniquely determine the target, split, preprocessing, or hyperparameters. Conversely, a faithfully specified method may fail for an ordinary engineering reason and never yield a comparable number. Treating these outcomes as one binary label—"reproduced" or "not reproduced"—hides the evidence needed to interpret a forecasting benchmark.

This distinction is consequential in urban electric-vehicle (EV) charging forecasting. Open resources now cover zonal charging demand in Shenzhen, station occupancy in Paris, workplace charging sessions, and harmonized measurements across cities and continents [1–5]. These resources do not define interchangeable prediction problems: occupancy rates, port states, session records, charging duration, and energy volume have different denominators, missingness mechanisms, and operational meanings. At the same time, the associated modeling literature spans physics-informed graph networks, language-model formulations, exogenous-variable Transformers, linear decompositions, and pretrained forecasters

[6–11]. A reported score therefore becomes interpretable only after the data semantics, information set, forecast origin, and evaluation protocol are fixed.

Forecasting research has long emphasized out-of-sample and rolling-origin evaluation [12, 13]. More recent guidance documents leakage through preprocessing, inappropriate random splits, weak benchmarks, and metric choices that do not match the data or decision problem [14, 15]. Reproducibility programs likewise encourage code, data, environment, and reporting artifacts [16]. These practices are necessary, but a remaining gap is operational: once a large matrix has been partially recovered and executed, how should a researcher distinguish disclosure quality from execution status, predictive opportunity from realizable gain, and a statistically detectable difference from an improvement large enough to justify added serving cost? Without explicit distinctions, additional model search can gradually convert an evaluation interval into a tuning resource.

We address this gap through an artifact-grounded case audit of one UrbanEV forecasting inventory and its subsequent model-development chain. The intended readers are benchmark authors, artifact evaluators, and reviewers who must decide which urban-forecasting claims are licensed by a heterogeneous experiment record. The audit freezes a six-state configuration taxonomy, verifies target and run identities, and follows each proposed improvement through a sequential evidence ladder. That ladder separates an infeasible oracle diagnostic, forecast-origin predictability, strict prequential fixed fusion, learned routing, synchronous latency, foundation-model teacher qualification, residual distillation, and development-only replication on a separately isolated Paris panel. Fail-closed metric locks prevent incomplete or invalid artifacts from entering aggregate results, while internally time-stamped minimum-effect gates determine whether a stage authorizes the next one. The contribution is an auditable claim grammar and workflow instantiated by this case, not evidence that the framework has already reduced false discoveries across independent teams.

The audit changes the apparent story. Of 1,224 frozen configurations, 865 have completed artifacts, but only 156 satisfy the joint exact-protocol-and-numerical criterion. In the internal UrbanEV rolling evaluation, a strictly fitted CAPER–TimeXer fusion reduces macro RMSE by 3.178% relative to the strongest evaluated single model, yet increases batch-1 median synchronous latency by 48.86% on the frozen laptop-GPU workload. A learned router adds only 0.443% over that fixed reference and therefore fails the internally time-stamped 1% progression gate despite a positive block-bootstrap interval. The same discipline rejects Chronos-2 as the project teacher under the exact UrbanEV protocol, stops residual distillation after a 0.119% gain over ground-truth-only training, and withholds student training on Paris when a strict development-only teacher improves by 0.847% rather than the required 1%. These are scoped gate outcomes, not general verdicts about routing, foundation models, or distillation.

We organize the study around four research questions:

RQ1 How much of a large forecasting configuration matrix is exactly reproducible, approximately recoverable, trend-only, underidentified, failed, or never completed?

RQ2 When two experts are complementary, how much of an infeasible oracle gap survives strict prequential fixed fusion and learned routing?

RQ3 How do apparent accuracy gains change once synchronous inference latency and a frozen minimum-effect threshold are enforced?

RQ4 How do fail-closed audits, control branches, independent evidence, and stopping rules change the claims that remain defensible?

This paper makes four contributions:

- **A configuration-level evidence taxonomy.** We separate protocol and numerical reproduction, disclosure-limited recovery, trend-only recovery, underidentification, execution failure, and non-completion for a frozen 1,224-configuration inventory. The resulting registry makes clear why execution completion is not a substitute for exact reproduction.
- **A portable audit framework and claim grammar.** We type evidence objects as configuration, prediction, target, metric, latency, or protected-storage artifacts; attach identity and lock requirements; and specify the narrow wording each evidence state authorizes. A claim-to-artifact ledger and reviewer checklist make the framework inspectable rather than leaving it as project history.
- **A sequential audit of model-improvement claims.** We connect target-identity checks, prequential fitting, control branches, block-resampled uncertainty, and hardware-specific latency to

explicit authorization gates. This exposes where theoretical complementarity becomes a material fixed-fusion gain and where later routing, teacher, and distillation signals remain below their frozen effect requirements.

- **A fail-closed evidence boundary.** We document audit events that blocked invalid shuffled controls and repaired missing-target representation before metric access, and we procedurally isolate Paris formal and protected target-bearing sources so that only development targets contribute analytically. The workflow produces an auditable mapping from each numerical statement to its protocol, artifact, and permitted wording.

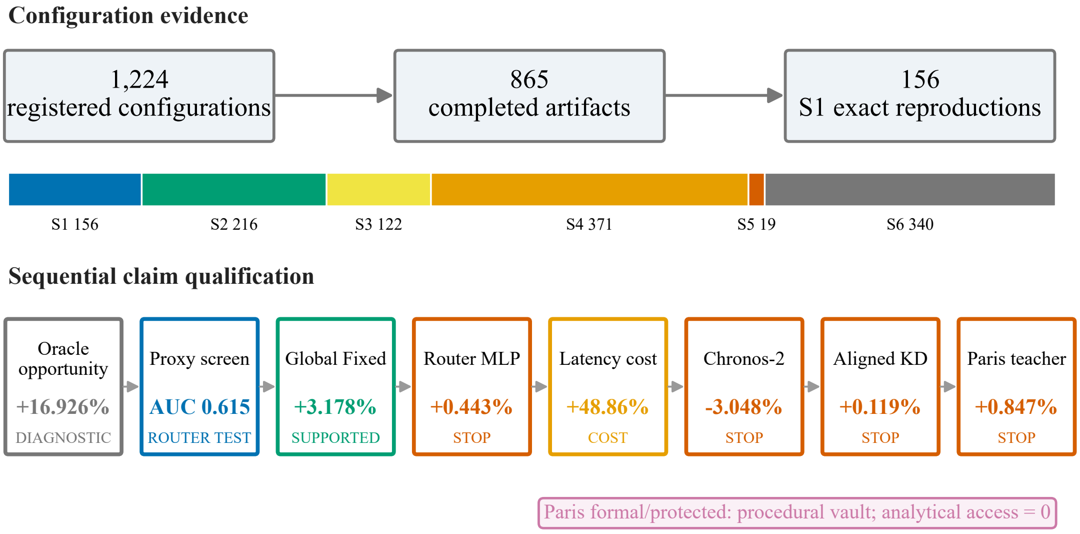

Figure 1: The audit separates registration, completed execution, exact reproduction, predictive opportunity, measured engineering cost, and qualification. A positive diagnostic or comparison does not automatically authorize the next stage; Paris formal and protected targets remain in procedural vault-only storage with zero analytical use.

Claim boundary. This is an empirical audit and evaluation study, not a new state-of-the-art model paper. Its numerical conclusions concern one UrbanEV inventory, six internal rolling folds at seed 42, one frozen latency platform, and Paris development-only evidence. UrbanEV and Paris absolute RMSE values are not compared because the panels and targets differ. The 1% threshold is an internally time-stamped project convention, not a universal definition of scientific or economic significance. Neither Paris formal nor Paris protected targets are accessed analytically, and we make no claim of protected-test performance, production readiness, universal generalization, or cross-dataset superiority. Their movement into independent vault-only storage is recorded as a storage event rather than concealed as non-materialization. Within that boundary, the case shows how configuration identifiability and sequential evidence controls constrain the claims this forecasting program can support.

The remainder of the paper first relates this audit to EV forecasting resources, time-series evaluation, and reproducibility research. It then defines the evidence taxonomy and sequential protocol, reports the registry and model-chain results, analyzes the fail-closed events and claim boundaries, and concludes with implications for auditable forecasting studies.

2 Related Work

EV charging data and prediction. Public EV charging resources differ in both observational unit and prediction target. ACN-Data records workplace charging sessions and supports behavioral and charging-control analyses [4]. UrbanEV instead provides hourly and finer-resolution zonal charging measurements for Shenzhen, together with prices, weather, points of interest, and spatial descriptors

[1]. CHARGED harmonizes charging duration, volume, price, weather, and geospatial attributes across six cities, making cross-city heterogeneity a central feature of the resource [5]. The Paris Smarter Mobility dataset records the evolving states of 273 charging points at 91 stations and was designed for occupancy forecasting at station, area, and global aggregation levels [2, 3]. These datasets broaden empirical access, but their targets are not semantically exchangeable. In particular, a session, an hourly charging-duration aggregate, a zonal occupancy ratio, and a non-available-port rate require different denominators and masks. Our study therefore audits target identity and uses each dataset only within its own protocol rather than pooling absolute errors across them.

Target heterogeneity also reflects distinct operational questions. Reviews of open EV load data distinguish session arrivals, connection and charging durations, energy, station load, and occupancy, while the broader forecasting literature addresses infrastructure siting, capacity planning, grid operation, and charging schedules with model-based, probabilistic, and machine-learning approaches [17, 18]. Forecasts may also be required at several aggregation levels and as distributions rather than point estimates: Buzna et al. develop an ensemble for hierarchical probabilistic station-load forecasting that uses information across lower- and higher-level series [19]. This operational lineage motivates treating the target quantity, aggregation level, forecast form, and decision horizon as protocol fields. We use it to delimit valid comparisons, not to claim a new probabilistic or hierarchical forecasting method.

EV forecasting methods have evolved in parallel with these datasets. Physics-informed graph learning incorporates spatial structure and a price–demand prior into regional demand prediction [6], while ChatEV reformulates heterogeneous charging inputs as a text-to-text prediction problem [7]. TimeXer models endogenous temporal patches together with exogenous variables [8]; by contrast, DLinear shows why simple decomposition and linear projection remain necessary controls for long-horizon forecasting claims [9]. We do not attempt to rank these method families across papers. Their reported tasks, data releases, splits, and metrics differ, and our locally audited compact TimeXer and CAPER implementations are project-specific experimental identities. The relevant question for this paper is narrower: which comparisons remain valid after target, history, horizon, fold, and evidence source are made identical?

Time-series evaluation. The prequential view evaluates a sequence of forecasts using information available before each outcome is observed [12]. Rolling-origin evaluation operationalizes this idea by moving the forecast origin through time, optionally refitting the model and producing multiple test periods [13]. Cross-validation for dependent data requires additional care: its validity depends on the prediction setup, stationarity, and residual dependence rather than on a generic rule that all time series must or must not use cross-validation [20]. Empirical comparisons find that temporally ordered out-of-sample estimators are especially appropriate under realistic non-stationarity [21]. Our strict prequential fusion and routing stages follow this lineage: the current evaluation fold never supplies labels for its own fusion weights or router training.

Evaluation design also determines what an error number means. Forecast metrics have different scale, denominator, and target-functional properties, so no single percentage error is reliable in every setting [22]. A broad review of common pitfalls identifies missing simple baselines, leakage through full-series preprocessing, mismatched horizons, and inappropriate aggregation as recurring causes of spurious forecasting conclusions [14]. Forecasting competitions further demonstrate both the value of strong cross-validation and combinations and the fact that complex machine-learning systems do not automatically dominate simpler references [23]. Our audit extends these principles from score computation to stage authorization: a positive difference must retain target identity, beat the strongest eligible reference, satisfy its frozen uncertainty and win-count conditions, and clear a separately declared minimum effect when the claim concerns engineering qualification.

Reproducibility and evidence auditing. Machine-learning reproducibility programs emphasize code, data, environment disclosure, and structured checklists as mechanisms for making empirical results inspectable [16]. Leakage reviews show why inspectability matters: information can cross a nominal train–test boundary through preprocessing, feature construction, temporal misalignment, or repeated analysis even when the final fitting call appears valid [15]. Benchmark results also vary with data sampling, initialization, and hyperparameter selection, and small differences can be dominated by those uncontrolled sources [24]. These findings motivate our run fingerprints, source hashes, target-array checks, and explicit separation between real-ground-truth, proxy, oracle, measured-latency, and artifact-integrity evidence.

Our stopping rules address a related form of adaptivity. A model-selection criterion can itself be overfit, creating optimistic performance estimates even when its underlying estimator is nominally unbiased [25]. Repeatedly querying a holdout also erodes its role as independent evidence; the reusable-holdout literature formalizes this problem for adaptively chosen analyses [26]. We use a procedural rather than privacy-based solution: freeze thresholds before aggregate access, lock metrics until artifacts close, rerun invalid controls non-selectively, and keep later formal and protected intervals physically separate. The contribution is not a new statistical test. It is an artifact-level account of how these principles alter an actual forecasting research program and its permissible claims.

Ensembling, foundation forecasters, and distillation. Forecast combinations often improve accuracy because component models retain different errors, and the M5 competition provides large-scale evidence for both simple and learned combinations [23]. Yet an oracle that selects the better expert after seeing the target is only an upper bound; it does not establish that forecast-origin information can recover the same gain. Our sequence from oracle complementarity to prequential fixed fusion and then learned routing makes that distinction explicit, while synchronous latency tests whether additional inference is justified by the retained accuracy improvement.

Pretrained forecasting models raise a parallel qualification question. Chronos tokenizes scaled numerical series and reports zero-shot probabilistic forecasting across diverse tasks [10]. Chronos-2 expands the formulation to grouped multivariate series and covariates through in-context interaction [11]. Such capabilities motivate direct evaluation, but a model's published aggregate ranking does not determine its role as a teacher under a different target, context length, covariate policy, and point-forecast rule. We therefore treat the locked Chronos-2 checkpoint as a candidate in the exact UrbanEV protocol rather than importing a reported score.

Distillation adds another layer of selection: teacher quality, student capacity, target alignment, and the ground-truth loss all affect whether teacher information improves the deployed student. In this audit, an aligned teacher branch is compared with both an identical ground-truth-only student and a shuffled-teacher control. Beating the shuffled control can identify sample-aligned signal without establishing a material gain over ground-truth training. This distinction connects the distillation experiment to the broader warnings about selection bias and benchmark variance [24, 25]. Accordingly, our related contribution is not a new distillation algorithm; it is a qualification protocol that stops teacher-cache or student expansion when the conjunctive gate fails.

3 Methods

3.1 Audit design and evidence hierarchy

We designed the study as an artifact-grounded audit rather than a search for a new state-of-the-art model. The unit of evidence was therefore not a headline score alone, but a linked record containing the protocol, configuration fingerprint, execution state, prediction and target identities, metric definition, and claim boundary. When records conflicted, we applied a frozen precedence rule: final decision and summary artifacts with a separately instantiated automated artifact audit took precedence over canonical result tables, which in turn took precedence over protocol thresholds and historical progress logs. Existing successful fingerprints were reused; failed or unexecuted configurations could only enter a later study under a new versioned identifier. This rule preserved the execution history and prevented a later successful run from rewriting an earlier failure or non-execution.

The audit separated four evidential questions. *Identifiability* asks whether the disclosed experiment determines a unique protocol and configuration. *Execution* asks whether that configuration reached a terminal run with an artifact. *Predictive evidence* asks whether predictions were evaluated against dataset targets under a time-respecting split. *Engineering qualification* asks whether the resulting effect passed a minimum-effect gate and its accompanying consistency checks. Proxy classification outcomes used to predict which expert would win locally were labeled separately from real-ground-truth forecast metrics and were never treated as forecast-accuracy evidence.

Table 1 is the portable object extracted from the case: each claim must name its evidence object, identity checks, lock or gate, and the strongest wording it authorizes. It is a proposed review instrument derived from one project, not an externally validated guarantee that other teams will reduce false discoveries by adopting it. The complete claim ledger and lock schemas are therefore reported in the Supplementary Material rather than hidden in an executor narrative.

3.2 Six-level configuration taxonomy

We reclassified a frozen inventory of 1,224 UrbanEV configurations into six mutually exclusive evidence states. The priority order was execution fact (S5 or S6), non-identifiability (S4), and then numerical agreement (S1–S3).

S1: protocol and numerical result identifiable and reproduced. The run covered all six folds, followed the official-compatible protocol, and each of the four disclosed metrics differed by at most 2% from its mapped reference.

S2: protocol broadly consistent, with disclosure differences. The run completed and remained above trend-only evidence, but used an audited correction or had a 2–5% numerical discrepancy.

S3: relative trend only. At least one core metric differed by more than 5%, or the disclosed comparison cell could not be covered completely.

S4: original information insufficient for a unique configuration. The paper omitted an identifying element, such as node identity, or an extended configuration could not be mapped uniquely to a disclosed paper cell.

S5: configuration identified but execution failed. The manifest fixed a configuration, but the registry recorded a failed terminal state.

S6: runnable configuration not actually completed. The manifest froze a runnable configuration that was skipped or never registered as a completed run.

S1–S4 can all possess completed artifacts; only S1 denotes exact protocol-and-numerical reproduction under these frozen rules. Classification was a deterministic implementation, not an inter-rater coding study. The necessary/sufficient tests, overlap priority, one configuration example per state, and exact reason fields are given in the Supplementary Material; no alternative tolerance was used to change the frozen counts. Thus, the taxonomy measures the evidence available in this UrbanEV inventory and does not estimate a field-wide reproducibility rate.

3.3 Data boundaries

UrbanEV internal rolling boundary. In this paper's audited UrbanEV task, the target is hourly zonal occupancy rate rather than every quantity in the broader dataset [1]. Occupied-port counts were divided by the frozen zone capacity; the resulting $4{,}344 \times 275$ panel was finite and required no target imputation. All same-protocol forecasting comparisons used a 168-hour history, horizons $\mathcal{H} = \{3, 6, 9, 12\}$ hours, six canonical expanding-month folds, and seed 42. Each model in a fold–horizon cell used the same target indices, timestamps, target values, and zone ordering. Fold target ranges, counts, prediction hashes, and fit sources are indexed in the Supplementary Material. This boundary is an internal rolling evaluation, not a blind protected test.

Independent Paris boundary. The second boundary used the Paris Smarter Mobility Data Challenge records [2, 3]. For station z and hour t, the development target was the *non-available-port fraction*,

$$y_{t,z} = \frac{C_{t,z} + P_{t,z} + O_{t,z}}{A_{t,z} + C_{t,z} + P_{t,z} + O_{t,z}}, \tag{1}$$

where A, C, P, and O denote Available, Charging, Passive, and Other counts. Passive and Other were grouped with Charging because the operational target asks whether a port is unavailable for a new user; it is not an energy-demand, session-volume, or pure actively-charging target. Fifty stations with

Table 1: Compact claim-control framework. Full schemas and wording grammar are provided in the Supplementary Material.

Evidence object	Required control	Strongest released claim
Configuration	protocol identity, disclosure mapping, execution state	registered, executed, failed, or S1–S6 evidence state
Prediction/target	index, time, entity, mask, and value hashes	internal/development metric for the named cell only
Diagnostic/proxy	target-dependent or proxy role stated explicitly	opportunity bound or predictability screen, not forecast gain
Engineering measure	device, precision, workload, batch, synchronization	measured latency/throughput on the frozen serving graph
Gate/repair	metric lock, conjunctive decision, parent/child hashes	gate pass/fail or artifact-integrity statement
Protected evidence	role, storage state, access definition and counter	procedural separation and zero analytical use after lock

a stable development denominator were locked, totaling 150 ports. The development matrix spanned 3,624 hourly timestamps from 3 July through 30 November 2020; 180,073 of 181,200 station–hour cells were observed (99.378%).

We split the remaining timeline into a 50-day formal role (1 December 2020–19 January 2021) and a 50-day protected role (20 January–10 March 2021). Formal- and protected-role target-bearing sources were storage-materialized only in a procedurally isolated vault outside the main project. Their file identities, row boundaries, and hashes were audited as storage metadata, while analytical access remained zero. Here, *analytical access* means reading a formal/protected target for metric computation, model selection, feature construction, statistical summary, or claim formation; hashing, moving, and inventorying bytes are storage operations and do not increment that counter. Two example rows were displayed during pre-lock schema discovery, before a model or selection rule existed, and are disclosed as a boundary limitation. No formal or protected target was used after the lock for a metric, model decision, or paper result. The separate storage migration is recorded as a storage event and is not described as absence of materialization.

Paris development models shared the same station order, targets, observation mask, history length, and horizons. Within each fold, missing inputs were filled causally: forward filling was limited to four hours, followed by a station-by-source-local-hour-of-week median fitted on the training interval, and then a training-station median. The observation mask was not a model feature, and training loss and evaluation used observed targets only. Because released timestamps did not encode UTC offsets, we used the unique source-naive ordinal-hour sequence; 24- and 168-hour lags therefore refer to ordinal data hours, not reconstructed UTC time.

The exact executed model identities, local adaptations, upstream-status limits, and source hashes are provided in the Supplementary Material and in `models/MODEL_SOURCE_MANIFEST.csv`.

3.4 Forecast metrics and strict prequential estimation

Let $\mathcal{O}_{f,h}$ be the observed target positions for fold f and horizon h, including all valid timestamps and spatial units. For method m, cell RMSE was

$$R^{(m)}_{f,h} = \sqrt{\frac{1}{|\mathcal{O}_{f,h}|} \sum_{(t,z)\in\mathcal{O}_{f,h}} \left(\widehat{y}^{(m)}_{f,h,t,z} - y_{f,h,t,z}\right)^2}. \tag{2}$$

The primary endpoint was the equal-weight mean of cell RMSEs,

$$\overline{R}^{(m)} = \frac{1}{|\mathcal{F}||\mathcal{H}|} \sum_{f\in\mathcal{F}} \sum_{h\in\mathcal{H}} R^{(m)}_{f,h}. \tag{3}$$

This cell-macro endpoint prevents a fold or horizon with more observed entries from dominating the paper's principal comparison. MAE, WAPE, sMAPE, and RAE were retained as secondary diagnostics.

For a candidate c and frozen reference r, positive relative gain denotes lower RMSE:

$$G(c;r) = 100\frac{\overline{R}^{(r)} - \overline{R}^{(c)}}{\overline{R}^{(r)}}. \tag{4}$$

Absolute UrbanEV and Paris RMSE values were not compared because the datasets differ in spatial unit, target construction, missingness, and fold definition.

We used strict prequential fitting [12, 13]: a parameter evaluated in fold f could only use validation data that preceded that fold or out-of-fold predictions from earlier folds. For the two UrbanEV experts, CAPER (C) and compact TimeXer (T), a fixed forecast was

$$\widehat{y}^{\text{fixed}}_{f,h} = \alpha_{f,h}\widehat{y}^{T}_{f,h} + (1-\alpha_{f,h})\widehat{y}^{C}_{f,h}, \qquad 0 \le \alpha_{f,h} \le 1. \tag{5}$$

The Global Fixed variant shared one α_f across horizons, whereas the Horizon Fixed control used four $\alpha_{f,h}$. Fold 1 used only pre-test validation predictions; folds 2–6 used only earlier-fold out-of-fold predictions. The target-dependent oracle selected the expert with smaller pointwise error and served only as an infeasible opportunity bound.

The expert-win classifier used 21 forecast-origin features whose timestamps did not exceed the forecast origin. Six described the experts (CAPER prediction, TimeXer prediction, signed and absolute disagreement, expert mean, and normalized horizon). Fifteen described the observed state: current occupancy, one-step change and its magnitude, daily and weekly deviations and their magnitudes, 24-hour mean, standard deviation and turnover, normalized capacity, and sine/cosine encodings of hour and weekday. Its AUC and Brier score quantified local expert-advantage predictability, not forecast accuracy or a causal mechanism. Learned routing was evaluated separately using real target RMSE. We compared a hard gate, a raw-probability soft gate, and a direct-loss multilayer perceptron; the direct-loss MLP was named as the primary learned-router comparison in the locked router protocol before aggregate Gate 4 access, even though the later raw-soft point estimate was numerically lower.

3.5 Uncertainty estimates and conjunctive gates

Paired uncertainty estimates resampled target timestamps in contiguous 24-hour blocks while keeping all spatial units at a timestamp together. The router used 1,000 bootstrap replicates. Distillation and Paris teacher qualification used 5,000 replicates. For Paris, the algorithm intersected timestamps across the four horizons within a fold, partitioned the ordered intersection into consecutive blocks, drew the original number of blocks independently with equal block probability, concatenated the selected timestamps, recomputed every cell RMSE, and then recomputed the equal-cell macro gain while reselecting the better single-model reference in that replicate. Each monthly sequence ended with a 12-hour trailing partial block; retaining it produced variable replicate length but equal expected inclusion for each original timestamp. We did not run post-hoc 48- or 168-hour alternatives after seeing the result. Intervals are therefore descriptive, protocol-specific paired 95% bootstrap intervals for relative gain; a positive interval did not override a failed minimum-effect gate.

Every qualification decision was conjunctive. The 1% point rule was an internal progression convention intended to decide whether added search, cache construction, student expansion, or protected evaluation was worth authorizing; it was not derived from charger-operator utility and is not a universal significance threshold. The relevant lock artifacts predate their respective metric unlocks and are identified by SHA-256 prefixes in the Supplementary Material; complete hashes are retained in the public machine-readable audit receipt. The displayed prefixes are router protocol `0b1fc56d...e7f18d35`, distillation protocol `fcd07f18...dfe383bc`, and Paris gate specification `96babeb1...90e9561`. The UrbanEV router had to beat TimeXer and both fixed controls, improve macro RMSE by at least 1%, win at least five of six folds and 18 of 24 cells, have a paired interval lower bound above zero, and avoid more than 1% deterioration at any horizon. Chronos-2 teacher qualification used the same 1%, fold, cell, and horizon criteria, together with target identity, causal input exposure, revision reproducibility, and cache feasibility. Residual distillation compared an identical 338-parameter student trained with ground truth only, aligned teacher residuals, or a 24-hour-block shuffled teacher. Passing required aligned KD to improve over GT-only by at least 1%, win five of six folds and 18 of 24 cells, have a positive paired lower bound, preserve every horizon within 1%, and beat the shuffled control in macro RMSE, at least five folds, and paired interval.

Paris teacher qualification was development-only. The strict prequential CAPER–TimeXer teacher had to improve over the replicate-wise best single model by at least 1%, win at least three of four folds and 12 of 16 fold–horizon cells, have a paired lower bound above zero, avoid more than 1% deterioration at any horizon, and pass target, station, mask, timestamp, and origin identity checks. A reported value below 1.000% failed the point-effect gate without rounding or threshold revision.

3.6 Teacher, distillation, and latency protocols

We screened the official `amazon/chronos-2` model [11] from a locked local snapshot at commit prefix `29ec3766`; the complete revision is retained in the model-lock artifact. The screen used `chronos-forecasting 2.2.2`, zero-shot FP32 inference, a 168-hour multivariate context, no future covariates, and the native median (Q0.5) forecast. Predictions were clipped to $[0, 1]$ for the frozen primary UrbanEV RMSE, matching the existing occupancy-rate evaluation.

The residual-distillation teacher was the audited Global Fixed fusion. The student combined a CAPER anchor with a shared DLinear residual branch [9]. GT-only, aligned-KD, and shuffled-teacher branches shared architecture, initialization, optimizer, epochs, data, and seed; only the teacher-residual term differed. This negative control tested whether any benefit required sample-aligned teacher information rather than merely an additional regularizer.

Synchronous latency used one RTX 4060 Laptop GPU in FP32 and a single-device sequential execution graph. One request represented a citywide refresh producing all four horizons for all 275 UrbanEV zones. Each method–batch combination used 50 warm-up iterations and 1,000 timed iterations in each of five independent processes; batch sizes were 1, 8, and 32 origins. Model-only time used CUDA events whose terminal event was synchronized. End-to-end wall-clock timing synchronized the GPU before stopping and included input preparation, transfers, four-horizon inference, fusion, and output transfer, but excluded model loading and checkpoint I/O, corresponding to a warm resident service rather than cold start. Batch-1 end-to-end median latency was the frozen primary engineering endpoint; the complete batch 1/8/32 table and software/hardware manifest are reported in the Supplementary Material.

3.7 Fail-closed artifact audit

The pipeline withheld evaluation metrics until each registered matrix closed and its predictions passed identity checks. An invalid control mapping, target mismatch, boundary violation, incomplete provenance, or representation defect blocked metric unlock. Corrections made under the lock were non-selective: a defective control family was rerun in full rather than only where an error was observed. Repair receipts recorded before/after hashes and whether predictions changed. Separately instantiated automated artifact audits were read-only checks against a frozen deployment contract; they could block execution or metric unlock but could not edit predictions or lower thresholds, and they were not external human replications. Canonical decision files were retained alongside negative results. The Supplementary Material formalizes the states, required lock fields, unlock conditions, and the weaker receipt-chain evidence that remains a limitation. Finally, the Paris storage guard separated storage materialization from analytical use: formal and protected target-bearing bytes remained in vault-only procedural storage, the main project contained no matching target source, and analytical access count remained zero.

4 Results

4.1 Completed execution was much broader than exact reproduction

The frozen registry contained 1,224 unique configurations with no duplicate run identifier or configuration fingerprint. Of these, 865 (70.67%) had completed artifacts, 19 had failed terminal states, and 340 had not been executed. Completion did not imply exact reproduction: only 156 configurations

(12.75%) satisfied S1, while 216 (17.65%) were S2, 122 (9.97%) were trend-only S3, and 371 (30.31%) were non-identifiable S4. Thus, the S1 count was less than one fifth of the completed-artifact count.

The distribution also depended on the original UrbanEV source table. UrbanEV source Table 3 contained all 156 S1 configurations, alongside 180 S2, 94 S3, six S5, and 44 S6 configurations. In contrast, 215 of 216 source Table 4 configurations were S4 under the frozen mapping rule. Source Table 5 contained 156 S4 and 296 S6 configurations among 528 entries. These patterns identify where disclosure and execution evidence was limited; they do not rank the predictive quality of the models appearing in those source tables.

4.2 Strong complementarity contracted under feasible fusion

The lightweight qualification matrix completed all 72 registered runs (three models, six folds, and four horizons). Among the evaluated single models, compact TimeXer had the lowest macro RMSE at 0.073709. DLinear reached 0.075362, NLinear 0.075711, and GBDT-Forecaster 0.078165; CAPER's existing same-target result was 0.075507. These comparisons qualify TimeXer only within the evaluated UrbanEV protocol, not as a universal strongest model.

CAPER and TimeXer nevertheless exhibited substantial local complementarity. Their mean residual Pearson and Spearman correlations were 0.8422 and 0.7638, respectively, and a target-dependent oracle reduced macro RMSE to 0.061233. This was a 16.926% improvement over TimeXer, but it was an infeasible upper bound because it used the realized target to choose the expert. The strict Global Fixed fusion recovered a smaller observed share of this opportunity: macro RMSE was 0.071367, 3.178% below TimeXer. It improved all six fold means and all 24 fold–horizon cells; horizon-level gains ranged from 2.890% to 3.645%. A paired block interval was not part of this earlier frozen accuracy-result protocol, so we do not present the result as having passed the later router/teacher gate. The Horizon Fixed control was nearly identical at 0.071372, confirming that post-hoc horizon-specific weights were not needed to obtain the fixed-fusion point result.

Predicting which expert would win was harder than the oracle gap suggested. A rolling classifier using expert outputs and disagreement alone had mean AUC 0.5370 (range 0.5227–0.5701). Adding 21 forecast-origin features increased LightGBM mean AUC to 0.6151, with minimum and maximum fold AUC of 0.5660 and 0.6682; mean Brier score was 0.2388. These proxy diagnostics justified a minimal router evaluation, but did not themselves establish a forecast improvement.

4.3 The learned router produced a positive but sub-threshold gain

The hard router underperformed Global Fixed, with macro RMSE 0.072759. Raw-probability soft routing obtained the lowest observed router RMSE, 0.070979, corresponding to a 0.543% gain over Global Fixed. The locked primary direct-loss MLP obtained RMSE 0.071050, a 0.443% gain; the better observed secondary point estimate did not replace that primary comparison. The MLP won five of six folds and 18 of 24 cells, improved every horizon, and had a paired 24-hour-block gain interval of [0.275%, 0.634%]. Its oracle-gap capture was only 3.123%.

All consistency checks therefore passed, but the 0.443% point gain did not meet the frozen 1% minimum effect. Gate 4 failed and router development stopped. The conclusion is not that routing lacked any signal; it is that the observed internal gain did not justify the added routing complexity under the registered rule.

4.4 The best UrbanEV accuracy carried a clear latency cost

The synchronous benchmark completed 90 of 90 process jobs and retained 90,000 raw timing rows. At batch one, TimeXer required 14.067 ms median and 15.952 ms P95 end-to-end latency. Global Fixed required 20.940 ms median and 33.123 ms P95. Thus, the 3.178% internal RMSE improvement came with 48.86% higher P50 latency and substantially heavier tail latency. NLinear and DLinear were much faster (0.674 and 1.166 ms P50) but had higher RMSE. Under this single-GPU sequential citywide-refresh workload, Global Fixed was the accuracy leader, not a low-cost ensemble. Batch 8/32, model-only, throughput, environment, and portability details appear in the Supplementary Material.

Table 2: Six-level evidence taxonomy in the frozen UrbanEV configuration inventory. Percentages use all 1,224 registered configurations as the denominator.

Code	Evidence status	Count	Share
S1	Protocol and numerical result identifiable and reproduced	156	12.75%
S2	Protocol broadly consistent; disclosure differences remain	216	17.65%
S3	Relative trend only	122	9.97%
S4	Original information cannot identify a unique configuration	371	30.31%
S5	Configuration identified; execution failed	19	1.55%
S6	Runnable configuration not actually completed	340	27.78%

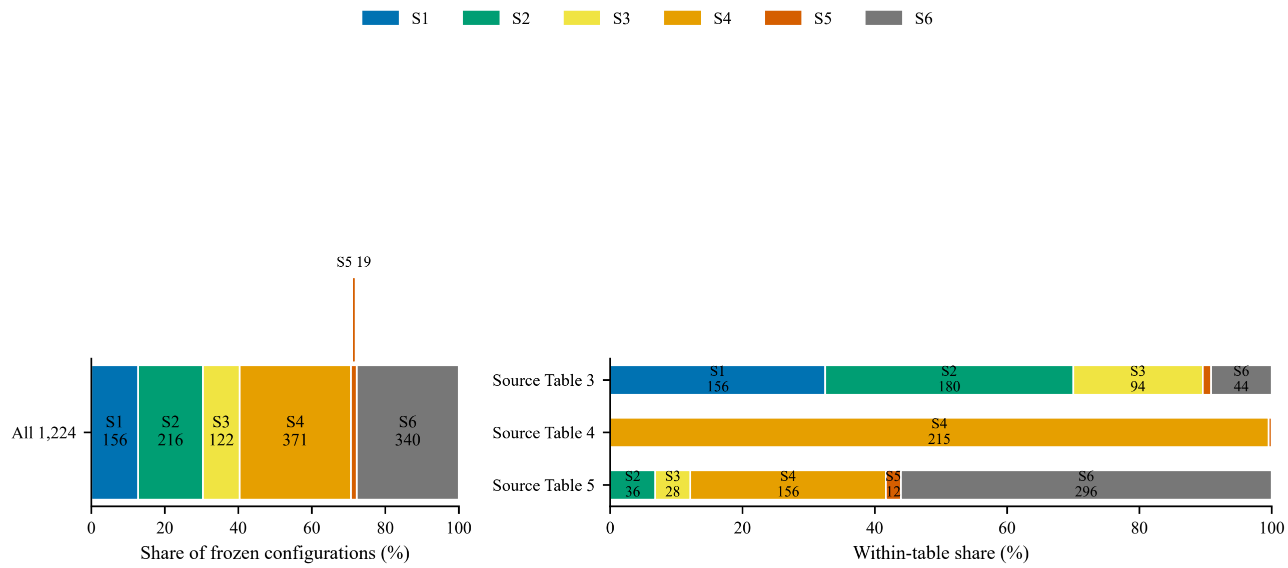

Figure 2: Evidence status in the frozen UrbanEV inventory, overall and by source table. Completed execution and exact reproduction are separate quantities; status counts do not measure intrinsic model quality.

4.5 Chronos-2 did not qualify as the UrbanEV teacher

The locked Chronos-2 screen completed all 24 fold–horizon cells and 5,960 forecast origins under the zero-shot, no-future-covariate, median point-forecast setting. Target, index, timestamp, and zone identities passed, all inputs ended no later than the forecast origin, and a repeated sentinel prediction had maximum absolute difference zero. Chronos-2 macro RMSE was 0.073542, compared with 0.071367 for Global Fixed, a relative gain of −3.048%. It beat the control in one of six folds and four of 24 cells; its RMSE deteriorated by 2.138%–3.932% across the four horizons. Although Chronos-2 had lower secondary MAE (0.042250 versus 0.044949), the pre-specified decision hierarchy used RMSE as the primary teacher endpoint and MAE as a secondary diagnostic. Chronos-2 therefore failed this exact teacher screen, and Global Fixed remained the audited teacher; this is not a general comparison of foundation forecasters with local models.

4.6 Aligned residual distillation exposed a weak teacher signal, not a stable gain

The final distillation matrix contained 72 audited cells: GT-only, aligned-KD, and shuffled-teacher branches across six folds and four horizons. Macro RMSE was 0.075568 for GT-only, 0.075478 for aligned KD, and 0.075674 for the shuffled control. Relative to GT-only, aligned KD improved RMSE by 0.119%, won four of six folds and 14 of 24 cells, and had a paired interval of [−0.064%, 0.303%]. It failed the frozen gain, fold, cell, and positive-lower-bound criteria, while horizon protection passed.

Aligned KD did outperform shuffled-teacher training by 0.259%, with five of six fold wins and a paired interval of [0.151%, 0.367%]. This contrast supports a small sample-aligned teacher signal, but not stable improvement over the identical GT-only student. The D1 gate failed, `distill_v1`

Table 3: Same-protocol UrbanEV accuracy results over six rolling folds and four horizons (24 equal-weight cells, seed 42). The oracle uses target-dependent expert choice and is not deployable.

Method	Cells	Macro RMSE	Macro MAE	Evidence role
GBDT-Forecaster	24	0.078165	0.049388	lightweight baseline
NLinear	24	0.075711	0.047894	lightweight baseline
CAPER	24	0.075507	0.048480	expert
DLinear	24	0.075362	0.048041	lightweight baseline
TimeXer	24	0.073709	0.046367	best evaluated single model
Global Fixed	24	0.071367	0.044949	strict prequential fixed fusion
Horizon Fixed	24	0.071372	0.044951	horizon-specific fixed control
Oracle	24	0.061233	0.035023	target-dependent upper-bound diagnostic

Table 4: Batch-1 end-to-end accuracy–latency trade-off on the frozen RTX 4060 Laptop GPU, FP32 single-device sequential workload.

Method	RMSE	P50 ms	P95 ms	Origins/s
NLinear	0.075711	0.674	1.056	1483.7
DLinear	0.075362	1.166	1.763	857.9
GBDT-Forecaster	0.078165	6.242	7.195	160.2
CAPER	0.075507	7.934	10.798	126.0
TimeXer	0.073709	14.067	15.952	71.1
Global Fixed CAPER+TimeXer	0.071367	20.940	33.123	47.8

terminated, and neither student latency nor multi-seed or protected evaluation was run.

4.7 Paris development repeated the distinction between consistency and qualification

Paris non-available-port-fraction results are reported only within the development boundary and are not compared numerically with UrbanEV. They are not energy-demand or session-volume forecasts. Among simple development baselines, last observation had macro RMSE 0.302630, daily lag 0.331654, and weekly lag 0.381137. In the 16-cell teacher qualification, the locally audited Paris CAPER adaptation obtained RMSE 0.244601 and the locally audited TimeXer adaptation 0.242542. The strict prequential fixed teacher reduced RMSE to 0.240488; the target-dependent oracle diagnostic was 0.217661.

The teacher improved over TimeXer by 0.8469%, won all four folds and all 16 cells, and had a paired interval of [0.6867%, 1.0088%]. Every bootstrap replicate selected TimeXer as the best single reference, and no horizon deteriorated. This is a consistent, near-threshold development result, but the point estimate remained below the frozen 1% progression rule. The conjunctive teacher gate therefore failed and the current Paris distillation hypothesis terminated before student training; it was not rounded into a pass and did not authorize formal/protected access.

Formal- and protected-role target-bearing sources existed only in the independent procedural vault. The storage boundary was materialized and hash-audited, but analytical access count remained zero; no post-lock target value or metric from either role contributed to the Paris result or this paper.

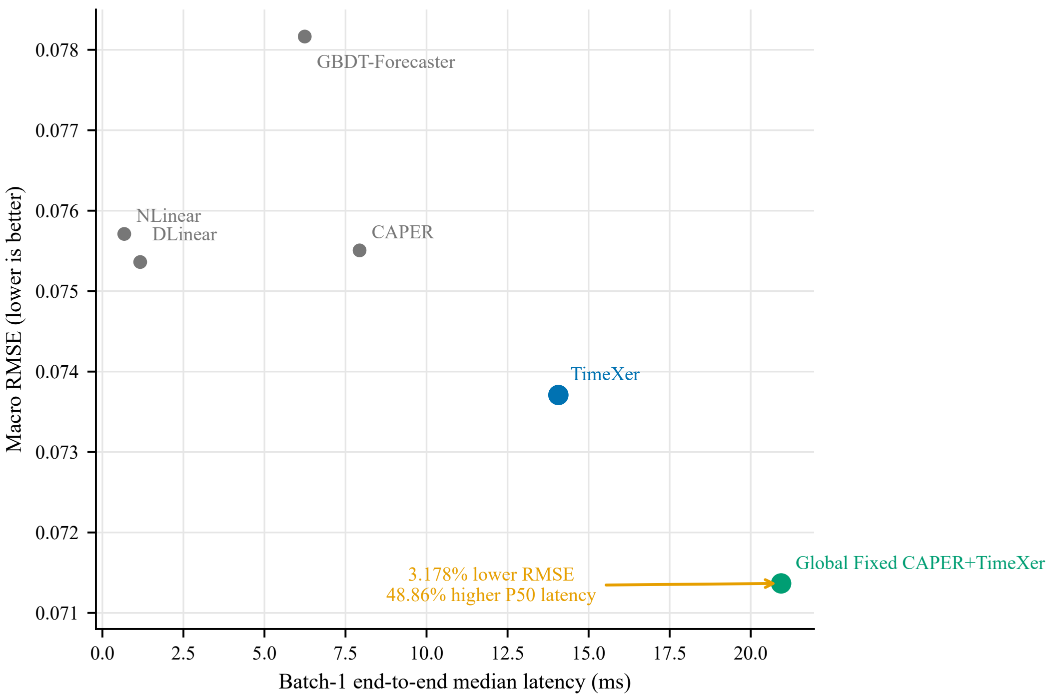

Figure 3: Batch-1 UrbanEV accuracy–latency frontier on the frozen RTX 4060 Laptop GPU in FP32. The arrow from TimeXer to Global Fixed represents 3.178% lower RMSE and 48.86% higher median end-to-end latency.

Table 5: Stage-specific qualification outcomes. Gains use different frozen references and are not absolute cross-stage or cross-dataset comparisons. "–" for Chronos-2 means that its frozen screen did not register a paired interval, not zero uncertainty.

Stage	Frozen reference	Gain	Paired 95% CI	Primary point gate	Decision
Router MLP	Global Fixed	+0.443%	[0.275, 0.634]%	1.0%	FAIL
Chronos-2	Global Fixed	-3.048%	–	1.0%	FAIL
Aligned KD	GT-only student	+0.119%	[-0.064, 0.303]%	1.0%	FAIL
KD vs shuffled	shuffled teacher	+0.259%	[0.151, 0.367]%	positive control contrast	PASS SUBCHECK
Paris teacher	Paris TimeXer	+0.847%	[0.687, 1.009]%	1.0%	FAIL

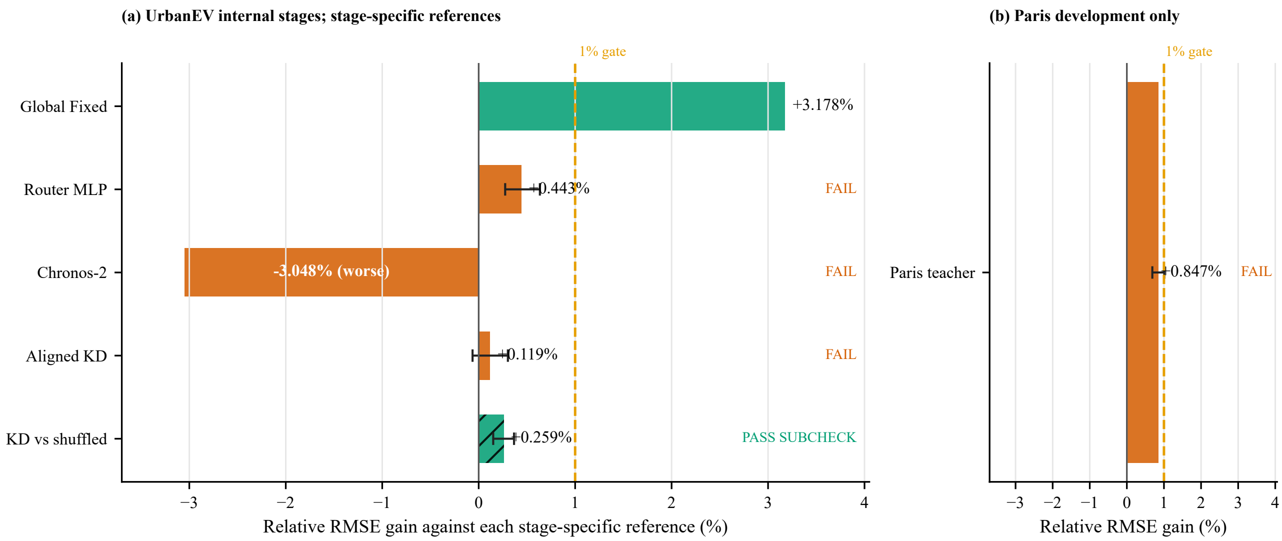

Figure 4: Stage-specific relative gains against each frozen reference. UrbanEV and Paris use different panels, targets, and references; the plot compares gate outcomes rather than absolute RMSE. The dashed 1% line is the project-specific minimum effect, not a universal utility threshold.

4.8 Fail-closed audits changed the admissible evidence before interpretation

The first M7-C matrix audit passed 69 of 72 cells and detected three shuffled-control mappings that were identities. Metrics remained locked. The pipeline invalidated and reran all 24 shuffled controls rather than selecting only the three detected cells; the final audit passed all 72 cells before D1 metrics were computed.

The Paris pipeline was similarly fail-closed. An initial separately instantiated automated artifact audit blocked deployment with eight implementation and evidence-boundary requirements. After the development-only shard, fingerprints, metric lock, synchronized bootstrap, provenance, and recovery rules were repaired, a final automated differential audit issued `DEPLOY_PASS`. All 32 model jobs then closed before metric access. A representation audit found fill values at masked target positions in 24 of 32 artifacts, covering 10,236 entries. The repair restored missing-value representation without retraining and left every prediction unchanged; metrics had not been accessed. A subsequent prediction audit passed 32 of 32 artifacts and then unlocked the development metrics. These automated audits were not external human replications.

These interventions support the integrity of the final negative decisions; they do not constitute evidence that auditing improved forecast accuracy. Throughout the Paris stage, formal/protected sources remained in vault-only procedural storage and analytical access count stayed at zero.

The four research questions are resolved by the linked configuration taxonomy, progression gates, accuracy–latency result, and fail-closed boundary evidence above; the complete claim-to-artifact map is provided in the Supplementary Material.

Table 6: Fail-closed interventions performed before scientific interpretation. Repairs did not change model predictions or optimize observed metrics.

Event	Detection	Required action	Final state
M7-C control	3 identity mappings detected (69/72 initial audit pass)	all 24 shuffled controls rerun	72/72 final audit PASS
Paris predeployment review	automated artifact audit: BLOCK	8 blockers repaired; DEPLOY PASS	pre-run authorization only; metrics locked
Paris target repair	24 runs; 10,236 masked targets restored	restore missing representation; predictions unchanged	metrics still locked
Paris metric unlock	32/32 jobs closed	prediction identity audit	prediction audit PASS; metrics unlocked; teacher gate FAIL; formal/protected analysis locked; analytical access 0

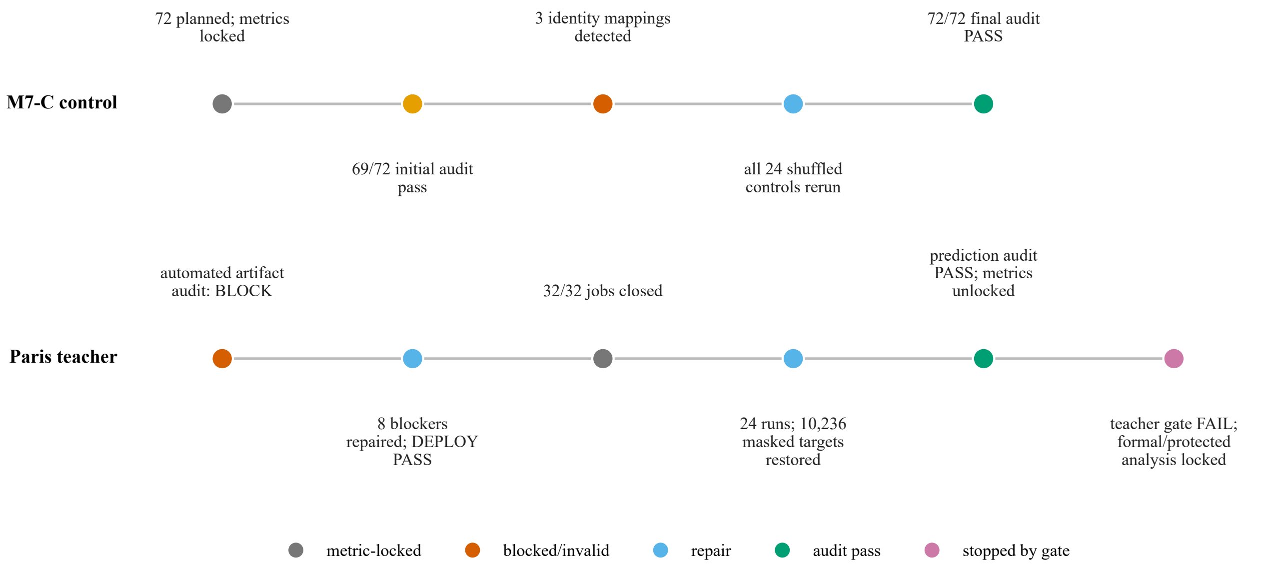

Figure 5: Fail-closed audit timelines. Invalid controls and missing-target representation were corrected before metric interpretation; the corrections did not alter model predictions or authorize formal/protected analysis.

5 Discussion

5.1 Completion is not reproduction

The configuration audit exposes a distinction that a conventional completed-run count would hide. Of the 1,224 registered UrbanEV configurations, 865 have completed artifacts, yet only 156 satisfy the frozen S1 criterion: the protocol is identifiable and all disclosed numerical results are reproduced within the prespecified tolerance. The remaining completed artifacts occupy qualitatively different evidence states. Some are broadly consistent but differ from the disclosure (S2), some recover only relative trends (S3), and others cannot be mapped to a unique source configuration (S4). A successful process exit can therefore establish that one implementation ran; it cannot establish that the implementation represents the reported experiment. This distinction also explains why configuration status should not be read as model quality. For example, the concentration of S4 entries in one source table reflects missing identification information, not an empirical finding that the associated models are weak.

This layered view is stricter than a binary "reproduced/not reproduced" label, but it is also more useful. It preserves partial evidence without promoting it to an exact-reproduction claim. The distinction is consistent with reproducibility frameworks that separate artifacts, procedures, and numerical outcomes [16, 24]. In our inventory, the practical consequence is large: reporting 70.67% artifact completion rather than 12.75% S1 reproduction would answer a different research question.

5.2 Oracle opportunity is not deployable gain

CAPER and TimeXer exhibit substantial local complementarity. A target-dependent oracle that selects the lower-error expert for each target reduces RMSE by 16.926% relative to the best evaluated single model. This value is informative as an upper-bound diagnostic, but the oracle uses the realized target and is therefore infeasible at forecast time. Once the decision must be made from information available at or before the forecast origin, the evidence contracts sharply. Strict prequential Global Fixed fusion retains a 3.178% internal RMSE improvement, whereas the learned direct-loss router adds only 0.443% over that fixed reference and recovers 3.123% of the oracle gap.

The gap between 16.926% and 0.443% is not a contradiction. It measures the difference between complementary errors that exist after outcomes are observed and complementary errors that can be exploited prospectively. The state-feature classifier's mean AUC of 0.6151 supports moderate predictability of relative expert advantage, not causal knowledge of charging demand and not equivalent forecast performance. This is why the evidence ladder separates oracle diagnostics, proxy-label prediction, real-ground-truth forecasting, and engineering qualification rather than treating them as interchangeable model scores.

Deployment relevance contracts the result once more. Global Fixed has the best audited UrbanEV internal RMSE, but synchronous batch-one inference on the frozen RTX 4060 Laptop GPU increases end-to-end P50 latency from 14.067 ms for TimeXer to 20.940 ms, a 48.86% overhead; P95 rises from 15.952 to 33.123 ms. The fixed fusion is thus an accuracy result with a measured hardware-specific cost, not a cost-free or universally efficient ensemble. The observed accuracy–latency trade-off also shows why a predictive improvement and a deployment recommendation require different evidence.

5.3 A positive confidence interval is not a frozen qualification decision

Several stages produced positive signals without satisfying their internally time-stamped gates. The router improved on Global Fixed by 0.443%, with a paired 24-hour-block interval of $[0.275\%, 0.634\%]$. The interval is entirely above zero, but it is entirely below the frozen 1% minimum-effect threshold. The Paris strict teacher provides an even sharper example: it won all four development folds and all 16 fold–horizon cells, and its interval was $[0.687\%, 1.009\%]$, yet its point improvement was 0.847%. The conjunctive rule required a point improvement of at least 1%, so the teacher did not qualify. The aligned distillation branch was weaker still: its 0.119% improvement over the identical ground-truth-only student had an interval of $[-0.064\%, 0.303\%]$ and failed the frozen fold and cell requirements.

These decisions do not assert that effects below 1% are scientifically meaningless. The threshold is a local progression convention, not an operator utility model or a universal economic or statistical constant. Rather, the decisions preserve the interpretation fixed before result inspection. A confidence interval evaluated against zero addresses evidence for a signed effect; an engineering gate addresses whether the observed effect is large and consistent enough to justify the next stage. Treating the former as automatic satisfaction of the latter would silently replace the registered question after observing the answer. As post-hoc context only, a 0.5% point rule would still reject the router, Chronos-2, and aligned KD but would change the Paris development decision; a 2% rule would reject every downstream candidate. These counterfactuals do not revise the frozen 1% outcome.

5.4 Stopping rules make negative evidence reusable

The stopping rules prevented small positive results from initiating open-ended searches on the same evidence interval. Router v1 stopped after its effect-size failure. Chronos-2 remained an auditable zero-shot teacher screen rather than motivating revision-specific tuning after its −3.048% gain against Global Fixed. Distillation v1 terminated after the aligned branch failed its main gate, even though it beat the shuffled control by 0.259%. The current Paris teacher hypothesis likewise stopped on development rather than opening formal or protected targets to rescue a near-threshold point estimate. These stops protect future evidence boundaries and make the negative results interpretable: each failed branch answers a frozen question instead of becoming one iteration in an unreported adaptive search [25, 26].

Fail-closed controls reinforced the same discipline at the artifact level. In the distillation matrix, three identity mappings invalidated the intended shuffled controls while metrics remained locked. The audit invalidated and reran all 24 shuffled-control cells non-selectively, then unlocked metrics only after the final 72/72 artifact audit passed. In Paris, a separately instantiated automated predeployment artifact audit first blocked execution on eight issues. After correction, the 32 prediction artifacts closed without evaluation metrics; a later representation audit restored 10,236 masked target entries to missing values across 24 runs without changing predictions, and metrics remained inaccessible until all 32 artifacts passed the identity audit. These interventions support integrity of the final evidence. They do not support a claim that repair improved model accuracy.

5.5 The vault is an evidence boundary, not a performance result

Paris formal and protected target-bearing sources now reside outside the main project in a procedurally isolated evidence vault. The corrected inventory contains 48 non-self-referential files, including 29 hidden Git files and the object pack that could otherwise reconstruct removed ground-truth sources. Every manifest entry matches its recorded byte count and SHA-256 hash. Formal and protected storage shards cover 2020-12-01 through 2021-01-19 and 2021-01-20 through 2021-03-10, respectively. Storage materialization is recorded separately from analytical access: one migration event occurred, while the protected analytical-access count remains zero. The failed-attempt partial files are also inventoried as target-bearing evidence and were not analyzed.

This boundary supports a narrow but important claim: no post-lock Paris formal or protected target contributed to model selection, metric computation, or a paper result. It does not imply that the bytes never existed, that two pre-lock schema rows were never visible, or that Paris development is blind-test evidence. The protected assets include raw sources, reference files, Git objects, final shards, and failed partials. In-scope failures are accidental analysis, stale-cache reuse, path leakage, and recoverable Git history; a malicious authenticated user, operating-system compromise, and cryptographic confidentiality are out of scope. The vault is a path, manifest, hash, and process boundary with inherited filesystem permissions, not encrypted storage or hardened access control. Its value is epistemic: it makes the declared absence of analytical use inspectable and preserves those targets for a future versioned protocol.

5.6 Implications for EV charging forecasting studies

The case yields positive guidance rather than only prohibitions. Authors should name whether a target represents energy, sessions, active charging, occupancy, or port non-availability; report denominators and observation masks; separate model-search intervals from future operational evaluation; and tie latency to a concrete request shape. Reviewers should ask whether a completed run is still underidentified, whether an oracle or proxy has been promoted to forecast evidence, and whether a positive interval silently replaced a frozen minimum-effect question. Operational follow-up should complement macro RMSE with stakeholder-specific measures such as peak-period error, high-occupancy recall, calibrated intervals, or denied-service proxies for driver-facing systems. This paper does not estimate those utilities.

The portable author/reviewer checklist is reported in the Supplementary Material so that it can be reused independently of this case narrative.

6 Limitations

Scope of the registry. The 1,224-configuration taxonomy describes one frozen UrbanEV reproduction inventory. Its S1–S6 assignments depend on the available source disclosure, deterministic mapping rules, and internally fixed 2% and 5% numerical tolerances. The classifications were not independently double-coded, and no inter-rater reliability estimate is available. The resulting 12.75% S1 rate is not an estimate of reproducibility across urban forecasting as a field. Status counts also measure evidence identifiability and execution state, not intrinsic model quality.

Framework validation. The proposed audit framework is extracted from a single self-audited project. Separately instantiated automated artifact audits challenged specific protocol and result packages, but they were not external human replications, and no external team has applied the full claim grammar, state machine, or S1–S6 decision tree to a separate benchmark. The paper therefore offers a reusable instrument and worked case, not evidence that the instrument improves decisions or reduces false discoveries across laboratories.

Internal and development-only evaluation. The UrbanEV comparisons use six internal rolling folds, four horizons, and seed 42. Paris teacher qualification uses four development folds over a different panel, non-available-port target, missingness mask, and local audited model adaptations. Absolute UrbanEV and Paris RMSE values are therefore not comparable. The windows also differ behaviorally: Shenzhen covers September 2022–February 2023, whereas Paris development covers July–November 2020, when pandemic-era mobility, policy, infrastructure, pricing, land use, and fleet composition could differ. The evidence controls protocol leakage, not those transport-domain shifts. No UrbanEV protected evaluation was opened, and no post-lock Paris formal or protected target was used analytically. The results do not establish blind-test generalization, robustness across seeds or cities, production readiness, or state of the art.

Restricted candidate and implementation set. The baseline conclusion applies only to the evaluated candidates. The Chronos-2 screen fixes one official model revision, zero-shot FP32 inference, and no future covariates; it does not establish that foundation models generally underperform on charging data. Likewise, the CAPER and compact TimeXer Paris implementations are local audited adaptations rather than claims about every upstream implementation. The residual-distillation result concerns one 338-parameter student and one seed, and the stopped protocol did not authorize a student architecture search, multi-seed expansion, or latency evaluation.

Oracle, proxies, and uncertainty. The oracle is target-dependent and only bounds latent complementarity. The expert-advantage classification task uses a proxy label derived from relative expert errors; its AUC does not prove causal predictability or forecast improvement. Paired block-bootstrap intervals account for temporal dependence under the chosen 24-hour blocking scheme, but they do not eliminate all uncertainty from fold construction, model selection, or a small number of seeds. Each Paris fold retained a trailing 12-hour partial block, and no post-hoc 48- or 168-hour sensitivity was run, so its interval should be interpreted under that exact resampling implementation. The 1% minimum-effect gate is a project-specific progression rule without an accompanying economic utility model, and it should not be generalized as a universal significance threshold.

Hardware-specific latency. Latency was measured in FP32 under single-device sequential execution on one RTX 4060 Laptop GPU; batch one was the frozen primary endpoint, while batches 8 and 32 are descriptive appendices. A request produced predictions for 275 zones and four horizons. Kernel fusion, parallel expert execution, quantization, compilation, batching, cold start, a different accelerator, or a serving system could change both the absolute timings and the fusion overhead. The measured 48.86% P50 overhead is reproducible for the frozen warm-resident workload, not portable to arbitrary deployments.

Audit coverage. Fail-closed checks detected identity shuffles, target-representation errors, incomplete provenance, and premature metric access. These checks reduce known integrity risks but cannot prove the absence of every implementation defect. The repairs were verified through hashes, identities, and unchanged predictions; they were not independently reimplemented from scratch. The automated Paris result artifact audit retained non-decisive warnings: the teacher-weight CSV did not carry complete upstream provenance, the pre-unlock state lacked a separate hash snapshot, and the receipt-update chain was weaker than the prediction-identity audit itself. None can reverse a point estimate below the frozen gate, but they limit stronger process claims. Two example source rows were displayed during early schema discovery before the Paris boundary lock, when no model or selection rule existed. This event was logged and was not used for model choice, but it prevents an absolute claim that no target value was ever visible before boundary creation.

Dataset-license metadata. The Paris challenge repository and the associated Zenodo record expose different license labels (ODbL and CC BY 4.0, respectively) [2, 3]. This paper treats them as source-specific metadata and does not resolve the legal relationship between the records. Downstream redistribution should therefore verify the applicable terms at the exact source used.

Vault boundary. The final vault manifest covers 48 non-self-referential files, including 29 Git files, and all recorded hashes verify. Analytical access to protected labels is zero, but storage migration materialized formal and protected shards and retained redundant raw, reference, final-shard, and failed-partial copies. The partials were inventoried but not analyzed. The vault inherits filesystem permissions, including modify rights for authenticated users; it is procedural isolation rather than encryption or hardened authorization. Some truth-derived tutorial figures and score summaries remain outside the exact-target isolation set because they cannot reconstruct station-level labels. These properties limit the claim to verified source separation and zero analytical use.

Negative-result interpretation. A failed frozen gate does not prove that an entire method family cannot work. It establishes only that the audited candidate, reference, data interval, and qualification rule did not authorize progression. Conversely, a passed subcheck—such as aligned distillation outperforming a shuffled teacher—does not upgrade the failed main comparison. Our conclusions concern the evidential status of these experiments, not universal impossibility.

7 Conclusion

This audit separated four questions that are often collapsed in large forecasting studies: whether a configuration is identifiable, whether an implementation completed, whether it carries predictive signal, and whether the resulting gain is large and reliable enough to justify progression. In the frozen UrbanEV registry, 865 of 1,224 configurations have completed artifacts, but only 156 meet the exact-protocol-and-numerical S1 criterion. Completion is therefore useful evidence, but it is not reproduction.

The same separation changed the model narrative. A target-dependent oracle exposed 16.926% complementarity, while strict prequential fixed fusion retained a 3.178% internal gain and incurred a 48.86% batch-one P50 latency overhead on the frozen device. Learned routing, Chronos-2 teacher qualification, aligned residual distillation, and the Paris development teacher each answered their registered questions, but none passed its frozen progression gate. In particular, statistically positive router and Paris intervals did not override the project-specific 1% minimum-effect requirement.

Fail-closed metric locks, non-selective control reruns, prediction-identity audits, and internally time-stamped stopping rules kept these negative results interpretable. The procedural Paris vault further separates future formal/protected evidence from the present manuscript: 48 non-self-referential files, including 29 recoverable Git files, are hash-inventoried outside the main project, while protected analytical access remains zero. This is a workflow boundary, not encrypted storage and not a protected-test result.

The central conclusion is methodological rather than architectural. In this case, separating configuration identity, execution, statistical evidence, practical effect, latency, and protected-data use makes the supported scope inspectable. The accompanying claim grammar is a reusable proposal, not yet a cross-team validated instrument. Future work should apply a newly frozen model and loss specification to an untouched boundary, use multiple seeds and hardware environments, and independently test the same fail-closed progression rules. Until such evidence exists, the strongest supported outcome is an auditable internal accuracy–cost result accompanied by explicit negative gates and preserved future evaluation data—not a claim of state of the art or deployment readiness.

Data and artifact availability

The public repository contains versioned source snapshots, protocol/configuration locks, target-free prediction artifacts, machine-readable result summaries, and scripts that reconstruct the headline metrics after the user obtains the licensed source datasets. Paris formal and protected targets, raw reference files, recoverable Git objects, and private vault metadata are not distributed. Their public boundary receipt contains only role dates, hashes, row counts, and the zero analytical-access statement.

Funding

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Competing interests

The author declares no competing interests.

Ethics and scope statement

The study uses previously released charging datasets and contains no intervention on human participants. Charging observations may nevertheless encode mobility patterns, so the paper reports only aggregate forecasting evidence and does not attempt individual inference. No protected-test, deployment-readiness, or state-of-the-art claim is made.

AI assistance disclosure

Generative AI tools, including OpenAI Codex and ChatGPT, assisted with literature search, experiment-script development, artifact-consistency checks, separately instantiated automated artifact audits, figure preparation, and language editing. All reported numerical results were computed from stored predictions and licensed source targets by deterministic scripts. AI tools did not generate ground-truth targets, alter frozen thresholds, authorize formal or protected data access, or determine the final scientific claims. The confirmed first author reviewed the manuscript and accepts responsibility for its content.

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